# YASASVI NELLORE

## SOFTWARE ENGINEER

Dallas, TX | yasasvi.nellore@gmail.com | +1(636)252-6680 | https://www.linkedin.com/in/yasasvi-nellore/ | Github | Portfolio

### SUMMARY


- Full Stack & AI Engineer with 4+ years of experience designing scalable, distributed web applications and cloud-native microservices using Java, Python, and JavaScript/TypeScript.
- Strong expertise in microservices architecture, high-throughput backend systems, and API integrations across React, Spring Boot, and Node.js.
- Skilled in deploying cloud-native solutions with AWS, Docker, Kubernetes, and Jenkins, ensuring high availability, automation, and reliability.
- Currently focused on integrating AI and LLM-based systems (RAG, MCP, multi-agent orchestration, OpenAI/Gemini APIs) to enhance application intelligence and user experience.


### PROFESSIONAL EXPERIENCE

**TWISTED X – Global Brands** **TEXAS, USA**

Software Engineer – AI Dec 2024 - Present

- Built and maintaining an agentic AI supply chain intelligence platform using LangChain, LangGraph, and RAG, enabling multi-step reasoning, LLM orchestration, tool calling, eval harnesses, and retrieval across NetSuite ERP, supplier APIs, and logistics data, reducing reporting time by 70% and improving fulfillment by 15%.
- Engineered a web monitoring agent using Playwright and the Claude API to detect product listings across retailer sites, validate pricing compliance, and reduce false positives in matching workflows.
- Modernized inventory forecasting by replacing rolling 12-month historical averages (24–31% WAPE) with Google TimesFM 3 for time-series demand prediction, leveraging feature engineering and model fine-tuning to achieve 11–21% WAPE across SKUs and categories while reducing manual tracking by 40%.
- Designed scalable ETL pipelines using Python and Snowflake to ingest supplier, trading partner, and Locally sales data, standardizing disparate formats through data cleaning and transformation, and building partner-specific dbt marts for consistent reporting and analytics.
- Automated build, test, and deployment across platform services by streamlining CI/CD pipelines with GitHub Actions and Docker on AWS, enabling zero-downtime releases and cutting setup time from hours to minutes.
- Developed a product comparison tool using marqo-fashionSigLIP and DuckDB to match internal and competitor styles by image and attribute data, helping users find similar products, reduce duplication, and identify catalog gaps.
- Built MCP-based tool-calling agents through Celigo AI Studio, exposing NetSuite business logic via optimized RESTlet flows and JSON schema mapping, with guardrails, evaluations, rollback, and audit logging to ensure reliable execution while reducing retrieval latency by 40%.

**JPMORGAN CHASE** **TEXAS, USA**

Senior Associate - AI Research Jul 2024 - Dec 2024

- Contributed to early-stage AI research projects across Commercial & Investment Banking (Markets) and Asset & Wealth Management, partnering with data scientists and engineering teams to prototype and evaluate new approaches.
- Prototyped LLM-based solutions using retrieval-augmented generation and prompt engineering, improving response relevance by 15% over baseline on internal test sets.
- Formulated problems, generated hypotheses, and developed models in Python, running experiments and evaluations that improved baseline performance by 12% on target tasks.
- Built and tested machine learning pipelines in Python (PyTorch, scikit-learn, pandas) on large-scale financial datasets, delivering reproducible code and cutting data prep time by 35%.
- Strengthened service resilience with circuit breakers, API Gateway rate limiting, and OAuth2 security, supported by Prometheus and Grafana monitoring.
- Developed reusable Python utilities and evaluation scripts that standardized experiment tracking and reduced new-experiment setup time by 30%.

**COLOMBO AI** **TEXAS, USA**

Software Engineer – Intern Dec 2023 - May 2024

- Built ColomboAI's end-to-end personalized feed pipeline using TypeScript, Node.js, React, and PostgreSQL, generating relevant content from follows, interests, and user interactions, increasing engagement by 30%.
- Developed a multi-signal feed ranking engine and REST APIs using affinity, engagement, recency, and time-decay scoring, with cursor pagination, deduplication, and creator-diversity rules for personalized infinite scrolling.

- Implemented the creator posting experience, including multi-modal content tools ("Magic Pen") for images, video, and text, increasing content creation activity by 20%.
- Built a Node.js recommendation pipeline that captured user interactions in PostgreSQL, computed weighted affinity scores from likes, comments, and viewing behavior, dynamically re-ranked feeds using engagement, recency, and time-decay signals.
- Optimized PostgreSQL performance for high-volume social feed and messaging data by adding targeted indexes and restructuring inefficient queries, cutting average API response times by 40%.

**HCL TECHNOLOGIES** **HYD, INDIA**
Software Developer May 2021 - Jul 2022

- Delivered end-to-end Java/Spring Boot microservices on Agile teams, tuning SQL queries and integrating Elasticsearch to cut API latency by 300ms and save $12,000 in annual cloud costs.
- Built ETL and streaming pipelines processing 500 GB/day with Spark, Kafka, and HDFS, cutting job latency by 30% and improving data availability across downstream services.
- Developed 15+ responsive React UI components using TypeScript and Redux, optimizing rendering and Webpack code-splitting to reduce page-load time from 3s to 1.8s.
- Automated Jenkins CI/CD pipelines with Groovy and Docker, eliminating 10 hours of manual deployment weekly across 3 environments and enabling faster, more reliable releases.
- Migrated high-volume PDF and email workflows to AWS Lambda with API Gateway and CloudWatch, lowering EC2 runtime by 320 hours per year while holding p95 latency under 800ms.
- Refactored PostgreSQL and MongoDB schemas, cutting report generation time from 4.5s to 2.9s and increasing analytics throughput by 8,000 transactions per hour.

## SKILLS

- **Languages:** Java, Python, TypeScript, JavaScript, SQL, Go
- **Frontend:** React.js, Next.js, Angular, Redux, React Query, Webpack
- **Backend & APIs:** Spring Boot, Hibernate, FastAPI, Django, Node.js/Express, REST, GraphQL, Swagger/OpenAPI
- **AI & ML:** LangChain, LangGraph, LlamaIndex, Deep Agents,Scikit-learn, TensorFlow, PyTorch, Prophet, Hugging Face, Transformers, Diffusion Models, Prompt Engineering, Agentic RAG, MCP, Multi-Agent Orchestration, Tool/Function Calling, Semantic Routing, Embedding Models, Vector Search, LLM Evaluation, RAG Architecture, Model Monitoring, Drift Detection.
- **Big Data & Authentication:** Apache Spark, Hadoop, Hive, Pig, Scala, Kafka, Event-Driven Architecture, OAuth.
- **DevOps:** AWS, GCP, Azure, Docker, Kubernetes, Terraform, Jenkins, GitHub Actions, CI/CD
- **Databases:** PostgreSQL, MySQL, MongoDB, Oracle 11g, Redis, DuckDB, Data Modeling, Schema Design.
- **Monitoring & Tools**: Prometheus, Grafana, SonarQube, OpenAPI/Swagger, Postman, OpenTelemetry, Distributed Tracing, Logging, SLA/SLO Monitoring, Playwright
- **Testing & Methodologies:** JUnit, Selenium, Jest, Unit Testing, Integration Testing, End-to-End Testing, Mocking, TDD, Agile/Scrum, Secure SDLC, RBAC, PII Compliance/Masking, Deterministic Replay Testing, Idempotent System Design, AI Governance, Human-in-the-Loop

## EDUCATION

**UNIVERSITY OF TEXAS AT ARLINGTON** | USA Aug 2022 - May 2024
Master of Science in Computer Science, Specialization in Intelligent Systems and Big Data, Cloud Computing

## PROJECTS & RESEARCH

**AI-Mock Interview Platform | NEXT.js, Firebase, and Google Gemini LLM | Live Demo**

- Built an AI mock interview platform using Next.js, Firebase, and Google Gemini LLM, generating role-specific interview questions and follow-ups from user inputs.
- Implemented authentication, interview sessions, and AI-generated feedback, storing user responses and performance data in Firebase.

**Graduate Research Assistant**

- Created Python-based data processing pipelines, accelerating research analysis and improving accuracy of results.
- Assisted in building REST APIs for lab projects, enabling students and researchers to efficiently access and manipulate experimental datasets.
- Optimized backend performance by refactoring Java and Python modules, improving API speed and reliability.

**Tejas**

- Built a Python agent runtime with perception, retrieval, planning, execution, and memory for browser automation.
- Designed HITL approvals, policy controls, and audit logging for safe autonomous execution.
- Built a Playwright-based perception layer using accessibility trees and screenshots for agent planning.